

SymPy Bug Card

Bug 45: singularities misses real poles of a principal fractional-power denominator

Task. Find the real singularities of the reciprocal expression.

$$\frac{1}{(x^2)^{1/3} - 1}$$

SymPy's answer.

$$\text{singularities}\left(\frac{1}{(x^2)^{1/3} - 1}, x, \mathbb{R}\right) = \emptyset.$$

Correct answer.

$$\{-1, 1\}$$

Explanation. For real x , $x^2 \geq 0$, so $(x^2)^{1/3} = 1$ exactly when $x^2 = 1$. Substituting $x = 1$ or $x = -1$ makes the denominator zero, so both points are real poles.

Likely root cause. The diagnosis traces the failure through `singularities` into `solveset`: the real-domain rewrite produces an absolute-value equation, and `_invert_abs` rejects the whole finite target set $\{-1, 1\}$ because one branch is negative, discarding the valid $\text{Abs}(x) = 1$ branch too. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-045-singularities-misses-real-poles-of-a-principal-fractional-po/](#)